

LAYERMARK: MULTI-MESSAGE AUDIO WATERMARKING WITH FEATURE OVERLAP-AWARE EMBEDDING AND RECURSIVE DETECTION

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ABSTRACT

Audio watermarking has become an important technique for modern copyright protection by embedding identifiable information into audio signals. However, existing watermarking methods are primarily designed for single-message embedding, where newly embedded watermarks may overwrite previously embedded messages and lead to the loss of earlier watermark information. We observe feature overlap between different watermark messages as a key cause of this limitation. To address this limitation, we propose LayerMark, which introduces a Multi-Layer Adapter to reduce inter-layer feature overlap during embedding and a Multi-Layer Remover to progressively expose earlier embedded messages during detection. Experiments show that LayerMark detects messages across all four embedding layers while achieving an SNR of 33.62 dB. It also generalizes well to five unseen speech datasets, maintaining a macro-average PESQ of 4.137 and STOI of 0.988 after four sequential watermark embeddings, while demonstrating robustness to eight common audio transformations.

Index Terms—Audio Watermarking, Multi-Message Watermarking, Multi-Message Detection

1. INTRODUCTION

Audio watermarking plays an important role in copyright protection by embedding identifiable information into audio signals [1, 2, 3]. A modern audio watermarking system uses an encoder to transform a watermark message into audio features and embed them into the input audio. During verification, a decoder extracts the embedded features from the watermarked audio and reconstructs the watermark message [2, 3]. For example, AudioSeal [4] converts the watermark message into time-domain features and embeds them directly into the audio waveform, whereas Timbre [5], AudioMarkNet [6], and WavMark [7] convert the message into frequency-domain features and embed them into spectral representations of the audio.

Despite their different embedding representations, these methods typically embed different watermark messages into similar regions of the feature space. As a result, when a new watermark message is embedded into already watermarked

audio, its watermark message features overlap with the existing ones, causing overwriting the previous message. Consequently, if a non-owner re-embeds the audio using the same watermarking system but with a different watermark message, the newly embedded message may overwrite the original one, leading to provenance loss and ambiguity in ownership verification [8].

We find that this limitation arises because watermark message features generated for different messages are embedded into overlapping regions of the audio representation. Consequently, a newly embedded message can overwrite the features of the previously embedded message.

To address this limitation, we propose LayerMark. Unlike prior watermarking methods, LayerMark introduces a Multi-Layer Adapter to adjust newly generated watermark message features and reduce their similarity to previously embedded watermark message features, thereby reducing inter-layer overlap, together with a Multi-Layer Remover that recursively estimates and removes the currently visible watermark message feature during multi-layer detection, enabling earlier embedded messages to be progressively detected.

Our contributions are as follows:

- We propose LayerMark, a multi-message audio watermarking framework with a Multi-Layer Adapter for reducing inter-layer feature overlap and a Multi-Layer Remover for detecting earlier embedded messages.
- We demonstrate that LayerMark can embed and detect multiple watermark messages within the same audio. Compared with state-of-the-art watermarking methods, including Timbre, AudioSeal, and WavMark, which detect only the most recently embedded message.
- We evaluate LayerMark on perceptual quality and robustness, demonstrating its generalizability to unseen speech and robustness to common audio transformations.

2. RELATED WORK

Based on the embedding representation, existing audio watermarking methods can be broadly categorized into time-domain and frequency-domain approaches.

Time-domain Watermarking. Time-domain audio watermarking methods embed watermark information directly into the audio waveform. LSB-based [9] methods embed

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watermark bits by directly modifying the least significant bits of PCM audio samples. Echo-hiding [10] methods encode watermark information by adding low-amplitude delayed copies of the original audio signal, where different echo delays represent different watermark symbols. More recently, AudioSeal [4] generates learned time-domain watermark message features and directly adds them to the raw audio waveform.

Frequency-domain Watermarking. Frequency-domain methods first transform the audio signal into a spectral or transform-domain representation and embed watermark information in the transformed coefficients. Early methods employ spread-spectrum and frequency-hopping techniques [11] to distribute watermark information across frequency components. Timbre [5] embeds structured watermark patterns in frequency regions, while WavMark [7], SilentCipher [12], AudioMarkNet [6], RobustDNN [13] and IDEAW [14] adopt frequency-domain representations with different neural architectures for watermark embedding and detection.

Despite differences in embedding domains and architectures, existing methods primarily focus on single-message watermarking and do not support multi-message embedding.

3. METHODOLOGY

3.1. Overview

The overview of LayerMark is shown in Figure 1. The Watermark Generator $G_\phi(\cdot)$ converts the input message into watermark message features for embedding. The Detector $D_\psi(\cdot)$ detects the embedded message bits. When multiple messages are embedded, the Multi-Layer Adapter $A_\omega(\cdot)$ adjusts the newly generated watermark message features to reduce overlap with the previously embedded watermark layer. For multi-message detection, we adopt a layer-by-layer removal strategy. The Multi-Layer Remover $R_\theta(\cdot)$ estimates and removes the current watermark message feature, allowing earlier watermark messages to be detected. In the following sections, we describe how LayerMark reduces feature overlap during multi-message embedding and detects multiple embedded messages.

3.2. Model Architecture

We adopt the AudioSeal architecture [4] for the Generator $G_\phi(\cdot)$ and Detector $D_\psi(\cdot)$, and describe the Multi-Layer Adapter $A_\omega(\cdot)$ and Multi-Layer Remover $R_\theta(\cdot)$ below.

Multi-Layer Adapter. The Multi-Layer Adapter takes the newly generated watermark message feature and the watermark message feature from the previous layer as inputs. The concatenated features are processed by six 24-channel dilated convolutional blocks with dilation factors (1, 2, 4, 8, 16, 32). The output is divided into three branches for mask, gain, and correction estimation, which are combined to produce the adapted watermark message feature.

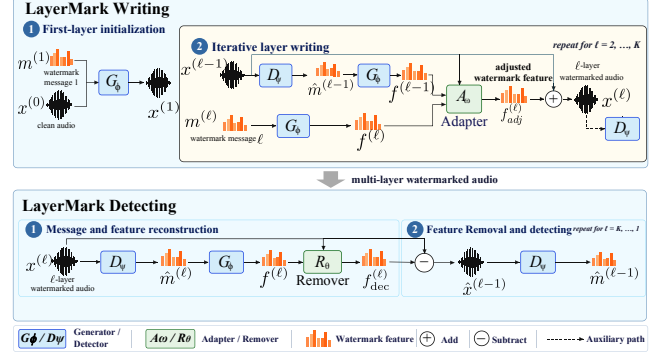


Fig. 1. Overview of LayerMark, including iterative multi-layer watermark writing and recursive watermark detection.

Multi-Layer Remover. The Multi-Layer Remover takes the watermarked audio and watermark message feature as inputs. The concatenated inputs are processed by eight 32-channel dilated residual blocks with dilation factors (1, 2, 4, 8, 16, 32, 64, 128). Two output branches estimate the gain and correction feature to produce the estimated watermark message feature.

3.3. Multi-Embedding Stage

Given a clean audio signal $x^{(0)}$ and K watermark messages $\{m^{(1)}, \dots, m^{(K)}\}$, LayerMark sequentially embeds the messages with $\ell = 1, \dots, K$ indexing the embedding order. At step ℓ , the Watermark Generator $G_\phi(\cdot)$ takes $x^{(\ell-1)}$ and $m^{(\ell)}$ as inputs and generates the watermark message feature $f^{(\ell)}$, producing the watermarked audio $x^{(\ell)}$. For $\ell = 2, \dots, K$, the Multi-Layer Adapter $A_\omega(\cdot)$ adjusts $f^{(\ell)}$ with respect to the preceding feature $f^{(\ell-1)}$, producing $f_{\text{adj}}^{(\ell)}$. We optimize message detection and perceptual quality at each step, with a feature-overlap objective.

Message Detection Loss. For the ℓ -th watermark layer, the Detector $D_\psi(\cdot)$ predicts $\hat{m}^{(\ell)}$ from $x^{(\ell)}$. During training, we apply the same audio distortions as AudioSeal [4] to $x^{(\ell)}$. The message detection loss is defined by comparing $\hat{m}^{(\ell)}$ with the ground-truth message $m^{(\ell)}$:

$$\mathcal{L}_{\text{msg}}^{(\ell)} = \text{BCE}(\hat{m}^{(\ell)}, m^{(\ell)}). \quad (1)$$

Layer Overlap Loss. To reduce overlap with the preceding watermark message feature, we encourage the adapted feature $f_{\text{adj}}^{(\ell)}$ to be less correlated with $f^{(\ell-1)}$. The layer overlap loss is defined using absolute cosine similarity:

$$\mathcal{L}_{\text{overlap}}^{(\ell)} = \left| \cos(f_{\text{adj}}^{(\ell)}, f^{(\ell-1)}) \right|, \quad (2)$$

Perceptual Loss. To preserve perceptual quality, we follow AudioSeal [4] and use three objectives: an L1 loss

$\mathcal{L}_{L1}^{(\ell)}$ to constrain watermark perturbations, a multi-scale Mel-spectrogram loss $\mathcal{L}_{\text{msspec}}^{(\ell)}$ [15] to preserve spectral characteristics, and an adversarial loss $\mathcal{L}_{\text{adv}}^{(\ell)}$ based on multi-scale STFT discriminators [16]. The overall perceptual loss is defined as

$$\mathcal{L}_{\text{perc}}^{(\ell)} = \lambda_{L1} \mathcal{L}_{L1}^{(\ell)} + \lambda_{\text{msspec}} \mathcal{L}_{\text{msspec}}^{(\ell)} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}^{(\ell)}. \quad (3)$$

Multi-Embedding Loss. The overall objective for the Multi-Embedding Stage combines the message detection, layer overlap, and perceptual losses across the watermark layers:

$$\mathcal{L}_{\text{emb}} = \sum_{\ell=1}^K \lambda_{\text{msg}} \mathcal{L}_{\text{msg}}^{(\ell)} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}}^{(\ell)} + \sum_{\ell=2}^K \lambda_{\text{overlap}} \mathcal{L}_{\text{overlap}}^{(\ell)}, \quad (4)$$

where λ_{msg} , λ_{overlap} , and λ_{perc} are weighting coefficients that control the contributions of the corresponding objectives.

3.4. Multi-Layer Detection Stage

After multiple watermarks are embedded, the Multi-Layer Remover $R_{\theta}(\cdot)$ starts from $x^{(K)}$ and recursively removes watermark message features layer by layer. At each step, the Detector $D_{\psi}(\cdot)$ detects the currently layer message, while $R_{\theta}(\cdot)$ estimates and removes its corresponding feature, enabling earlier messages to be detected. We define losses for feature reconstruction and message detection at each layer ℓ .

Watermark Message Feature Reconstruction Loss. The Multi-Layer Remover reconstructs the current watermark message feature by estimating $f_{\text{dec}}^{(\ell)}$ to match $f_{\text{adj}}^{(\ell)}$. We define the reconstruction loss using normalized mean absolute error (NMAE):

$$\mathcal{L}_{\text{rec}}^{(\ell)} = \text{NMAE} \left(f_{\text{dec}}^{(\ell)}, f_{\text{adj}}^{(\ell)} \right). \quad (5)$$

Audio Reconstruction Loss. After removing the estimated watermark message feature, the reconstructed audio $\hat{x}^{(\ell-1)} = x^{(\ell)} - f_{\text{dec}}^{(\ell)}$ should approximate $x^{(\ell-1)}$, the audio before the ℓ -th watermark was embedded. We therefore define the audio reconstruction loss by comparing $\hat{x}^{(\ell-1)}$ with $x^{(\ell-1)}$:

$$\mathcal{L}_{\text{audio}}^{(\ell)} = \text{NMAE} \left(\hat{x}^{(\ell-1)}, x^{(\ell-1)} \right). \quad (6)$$

Previous-Message Detection Loss. The Detector $D_{\psi}(\cdot)$ predicts the preceding watermark message $\hat{m}^{(\ell-1)}$ from the audio $\hat{x}^{(\ell-1)}$. During training, we apply the same audio distortions as AudioSeal [4] to $\hat{x}^{(\ell-1)}$ and define the detection loss as

$$\mathcal{L}_{\text{prev}}^{(\ell)} = \text{BCE} \left(\hat{m}^{(\ell-1)}, m^{(\ell-1)} \right). \quad (7)$$

Multi-Layer Detection Loss. The overall objective for the Multi-Layer Detection Stage combines the feature reconstruction, audio reconstruction, and previous-message detection losses across all watermark layers:

$$\mathcal{L}_{\text{det}} = \sum_{\ell=1}^K \left(\lambda_{\text{rec}} \mathcal{L}_{\text{rec}}^{(\ell)} + \lambda_{\text{audio}} \mathcal{L}_{\text{audio}}^{(\ell)} \right) + \sum_{\ell=2}^K \lambda_{\text{prev}} \mathcal{L}_{\text{prev}}^{(\ell)}. \quad (8)$$

where λ_{rec} , λ_{audio} , and λ_{prev} are weighting coefficients that balance the contributions of the corresponding objectives.

3.5. Total Training Loss

The overall training objective combines the losses from the Multi-Embedding and Multi-Layer Detection stages:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{emb}} + \mathcal{L}_{\text{det}}. \quad (9)$$

4. EXPERIMENTS

We conduct experiments to evaluate LayerMark in terms of multi-message detection performance, robustness to audio distortions, and perceptual quality.

4.1. Experimental Setup

Dataset Setup. The training set is drawn from LibriSpeech train-clean-100 [17], while the test set consists of AISHELL-1 [18] and VCTK [19]. For out-of-domain evaluation, we use AMI [20], FLEURS [21], ICSI [22], MediaSpeech [23], and Multilingual LibriSpeech [24].

Metric Setup. We use bit accuracy to evaluate multi-message watermark detection performance and report signal-to-noise ratio (SNR), PESQ [25], STOI [26], and ViSQOL [27] to assess the impact of watermark embedding on audio quality.

Baseline Setup. We compare LayerMark against four strong audio watermarking baselines: Timbre [5], AudioSeal [4], WavMark [7] and AudioMarkNet [6]. We evaluate all methods in multi-message watermarking settings, focusing on the successful detection of multiple embedded messages.

Embedding Layers and Training Parameters. We set the maximum embedding depth to $K = 4$. The loss weights are set to $\lambda_{\text{msg}} = 1.0$, $\lambda_{\text{rec}} = 0.25$, $\lambda_{\text{dir}} = 0.25$, $\lambda_{\text{dev}} = 0.05$, $\lambda_{\text{audio}} = 0.1$, $\lambda_{\text{prev}} = 1.0$, $\lambda_{L1} = 0.1$, $\lambda_{\text{msspec}} = 2.0$, and $\lambda_{\text{adv}} = 4.0$. For the overlap objective, we set $\lambda_{\text{overlap}} = 1.0$.

4.2. Multi-Layer Detection

LayerMark supports up to four watermark message embedding, while the other watermarking methods do not support multi-watermark detection. As shown in Table 1, Timbre, AudioSeal, WavMark and AudioMarkNet [6] detect only the latest message $m^{(4)}$, whereas LayerMark achieves accuracies of

0.66, 0.83, 0.94, and 1.00 for $m^{(1)}$ – $m^{(4)}$, respectively. LayerMark also achieves the highest SNR of 33.62 dB, demonstrating effective multi-message detection with better audio quality.

Table 1. Multi-layer detection accuracy and audio quality across different watermarking methods.

Method	$m^{(1)}$ Acc.	$m^{(2)}$ Acc.	$m^{(3)}$ Acc.	$m^{(4)}$ Acc.	SNR
Timbre	0.00	0.00	0.00	1.00	22.40
AudioSeal	0.00	0.00	0.00	1.00	20.43
WayMark	0.00	0.00	0.00	1.00	27.78
AudioMarkNet	0.00	0.00	0.00	1.00	25.12
LayerMark (Ours)	0.81	0.89	0.94	1.00	33.62

4.3. Generalization to Unseen Speech

LayerMark maintains strong perceptual generalization on unseen speech datasets. As shown in Table 2, E1–E4 denote sequential embedding of one to four watermark messages. As the embedding depth increases, the average SNR decreases from 26.48 to 22.24 dB and PESQ from 4.369 to 4.137, while STOI remains high at 0.988 for E4. Across the five unseen datasets, E4 achieves PESQ scores of 3.944–4.287 and STOI scores of 0.970–0.996.

Table 2. Perceptual quality on unseen clean speech with different numbers of embedded watermark messages.

Dataset	SNR ↑				PESQ ↑				STOI ↑				VisQOL ↑			
	E1	E2	E3	E4	E1	E2	E3	E4	E1	E2	E3	E4	E1	E2	E3	E4
AMI	25.08	23.10	23.70	20.59	4.291	4.171	4.051	3.944	0.985	0.980	0.975	0.970	4.884	4.824	4.779	4.734
FLEURS	25.73	23.87	22.60	21.62	4.257	4.145	4.062	3.984	0.994	0.992	0.989	0.988	4.912	4.858	4.818	4.783
ICSI	26.40	24.48	23.15	22.13	4.465	4.413	4.333	4.287	0.995	0.994	0.992	0.991	4.947	4.922	4.899	4.877
MediaSpeech	27.49	25.47	24.21	23.28	4.447	4.388	4.333	4.283	0.998	0.997	0.996	0.996	4.945	4.913	4.884	4.858
Multilingual LibriSpeech	27.71	25.79	24.51	23.56	4.387	4.323	4.257	4.185	0.997	0.996	0.995	0.994	4.914	4.871	4.832	4.798
Macro avg.	26.48	24.54	23.23	22.24	4.369	4.288	4.207	4.137	0.994	0.992	0.990	0.988	4.920	4.878	4.843	4.810

4.4. Robustness Evaluation

We test eight edits at fixed mild and moderate strengths [28, 29]: white noise (25/20 dB SNR), centered crop and padding (98/95% retained), low-pass filtering (7/6 kHz), MP3 (128/96 kbps), Opus (64/48 kbps), resampling (16 to 14 to 16 / 16 to

Table 3. Detection performance retained under audio editing across different watermark depths, normalized by matched-clean performance.

Edit	Mild				Moderate			
	$m^{(1)}$	$m^{(2)}$	$m^{(3)}$	$m^{(4)}$	$m^{(1)}$	$m^{(2)}$	$m^{(3)}$	$m^{(4)}$
Additive white noise	0.778	0.742	0.812	0.878	0.771	0.735	0.785	0.850
Centered crop and padding	0.765	0.786	0.863	0.902	0.758	0.772	0.859	0.899
Low-pass filtering	0.778	0.800	0.881	0.917	0.779	0.799	0.881	0.917
MP3 compression	0.534	0.476	0.526	0.597	0.511	0.475	0.526	0.597
Opus compression	0.794	0.802	0.880	0.918	0.812	0.802	0.880	0.918
Resampling	0.779	0.800	0.881	0.917	0.779	0.799	0.881	0.917
Real background noise	0.766	0.769	0.838	0.882	0.771	0.768	0.835	0.868
Temporal shift	0.500	0.493	0.512	0.642	0.756	0.778	0.860	0.904

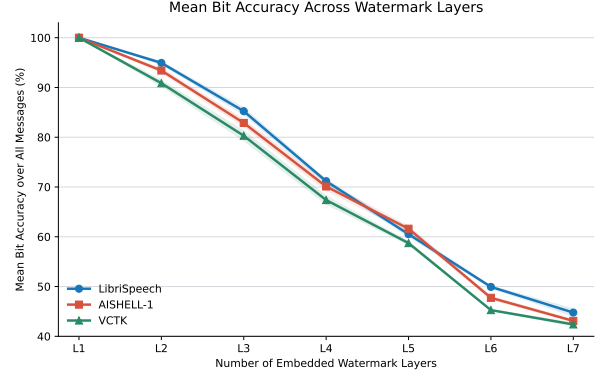


Fig. 2. Mean bit accuracy across different watermark embedding layers.

12 to 16 kHz), real noise (20/15 dB SNR) and temporal shift (+/-50 / +/-100 ms).

Results. Our watermark demonstrates robustness across different audio transformations and embedding layers. As shown in Table 3, later embedded messages generally retain higher detection performance under audio transformations. Under white noise, cropping, low-pass filtering, Opus compression, resampling, and real background noise, $m^{(1)}$ – $m^{(3)}$ retain approximately 0.74–0.88 of their clean performance, while $m^{(4)}$ retains 0.85–0.92. Low-pass filtering, Opus compression, and resampling are the most robust transformations, whereas MP3 compression and mild temporal shift cause the largest degradation.

4.5. Bit Accuracy Across Watermark Layers.

To study the effect of the number of embedding layers K on bit accuracy, we vary K from 1 to 7 and measure the mean bit accuracy over all embedded messages on LibriSpeech, AISHELL-1, and VCTK. We find that as the number of watermark layers increases, multi-message detection accuracy decreases, as shown in Figure 2. Mean bit accuracy drops from 100% at L1 to approximately 71%, 70%, and 67% at L4, and further to 45%, 43%, and 42% at L7 on LibriSpeech, AISHELL-1, and VCTK, respectively.

5. CONCLUSION

In this paper, we present LayerMark, a multi-message audio watermarking framework that reduces inter-layer feature overlap during embedding and enables recursive detection of earlier embedded messages. LayerMark introduces a Multi-Layer Adapter to adjust watermark message features across layers and a Multi-Layer Remover to progressively recover earlier messages. Experiments demonstrate effective detection across four watermark message embeddings, good perceptual quality on unseen speech, and robustness to common audio transformations.

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